

212.5 Gbps SerDes-Driven Optical Engines Based on SiPh and TFLN Platforms

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Abstract: We demonstrate a 212.5 Gbps SerDes-based LRO interconnect to compare SiPh and TFLN engines. We employ EIC-PIC analog peaking to overcome SiPh bandwidth limits. Conversely, while a SerDes direct-drive architecture improves TFLN's energy efficiency, it restricts signal integrity. This paper highlights these trade-offs. ©2026 The Authors

Introduction

Driven by the explosive growth in the parameter count of large language models, AI clusters demand high-bandwidth, low-latency, energy-efficient, and scalable interconnects to facilitate efficient training and inference [1-2]. Current data centers leverage high-throughput optical interconnects via pluggable modules to facilitate seamless connectivity between top-of-rack switches and diverse servers [3]. Within this network topology, high-performance SerDes lanes integrated with ASICs serve as physical-layer backbones, driving most of the cross-node data interactions [4].

In the development of optical interconnects, selecting an appropriate optical engine is crucial, as different platforms vary in bandwidth, loss, integration density, and cost [5]. Silicon photonics (SiPh) currently dominates due to its CMOS-compatible fabrication and high integration density [6]. However, its reliance on free-carrier dispersion imposes intrinsic bandwidth and modulation efficiency limitations [7]. Consequently,

improving the bandwidth and modulation depth of SiPh-based links remains a key research focus. In contrast, although thin-film lithium niobate (TFLN) modulators typically suffer from larger footprints and lower fabrication yield, they leverage the Pockels effect to achieve ultra-high electro-optic bandwidths exceeding 100 GHz with low drive voltages (< 0.8 Vpp), making them strong candidates for 400G/lane and beyond [8]. Moreover, effectively exploiting their low-voltage operation to enhance overall system energy efficiency also remains more exploration. As a result, the trade-offs between TFLN and SiPh platforms remain actively debated in the optical communications industry.

In this work, we improve the modulation depth and bandwidth of SiPh links through EIC-PIC co-optimization with an integrated driver, while adopting a direct-drive scheme to highlight the low-voltage advantage of TFLN. We experimentally demonstrate a 212.5 Gbps PAM4 optical interconnect based on an LRO architecture using a 212.5 G SerDes platform. Through a

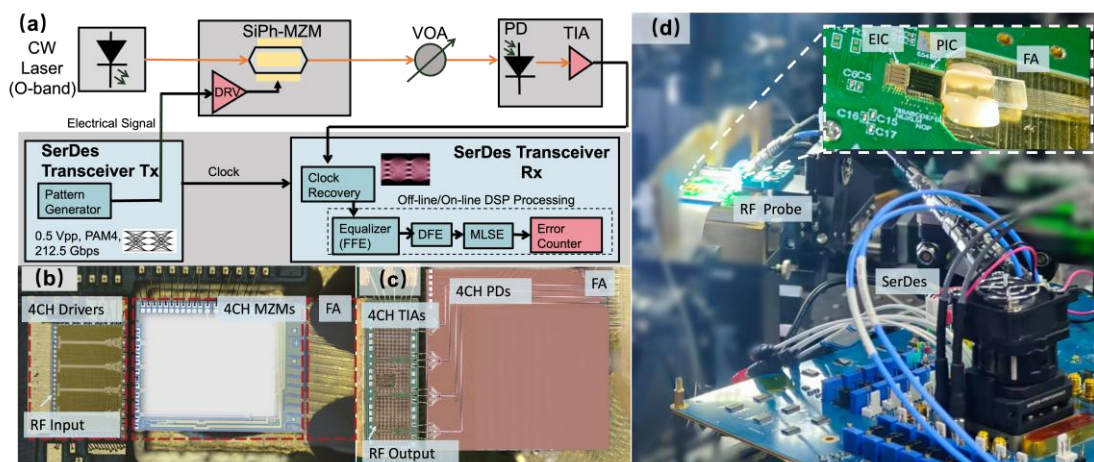


Fig. 1. SerDes-driven SiPh-based optical engine with integrated driver and TIA. (a) Schematic of the link architecture, (b) Microscope image of the integrated SiPh transmitter (EIC Driver and PIC MZM) and (c) receiver (PD and TIA arrays). (d) Experimental setup.

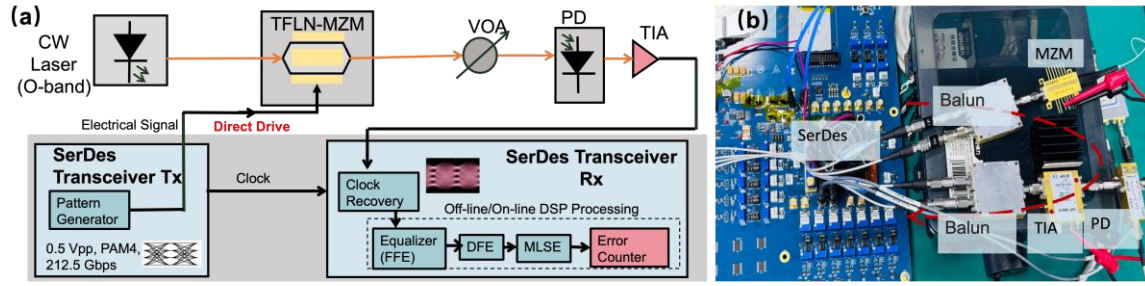


Fig. 2. SerDes direct-drive TFLN-based optical engine. (a) Schematic of the link architecture and (b) Experimental setup.

loopback link, we compare a driver-integrated SiPh engine with a driver-free TFLN engine. For SiPh, a DSP-free transmitter enabled by EIC-PIC co-optimization is validated: High-frequency analog peaking in the EIC driver is employed to compensate for the modulator's intrinsic 32 GHz bandwidth limit, enabling the SiPh engine to match the system-level performance of a direct-drive TFLN MZM in a 212.5 Gbps link. In contrast, the TFLN approach eliminates the external driver and relies on 0.5 Vpp SerDes direct drive, significantly reducing power consumption at the expense of reduced modulation depth. These results provide important experimental insights for platform selection in linear-driven single-wavelength optical interconnects for future AI clusters.

Experimental Setup

Figure 1(a) shows the constructed electrical-optical-electrical (E-O-E) loopback link. The SerDes

transmitter (Tx) generates PAM4 signals with adjustable peak-to-peak voltage swings (Vpp) as the signal source. The 10 dBm O-band continuous-wave (CW) laser serves as the optical carrier for modulators. At the receiver, the signal is detected by a photodetector (PD) and a transimpedance amplifier (TIA), then fed into the SerDes receiver (Rx) for DSP-based signal recovery and BER evaluation.

SiPh and TFLN Mach-Zehnder modulators (MZMs) are evaluated alternately. The SiPh transmitter (Fig. 1(b)) integrates a 4-channel MZM (EO bandwidth > 32 GHz, ~16.5 dB loss, 75 Ω impedance) and a 4-channel driver (53 GHz bandwidth, 20 dB gain, < 3 Vppd output). The receiver array (Fig. 1(c)) consists of 4-channel PDs and TIAs. The PDs exhibit a bandwidth of 90 GHz alongside a responsivity of 0.95 A/W, and the TIA provides a differential transimpedance gain of 8.7 k Ω and a low input-referred noise (2 μ A).

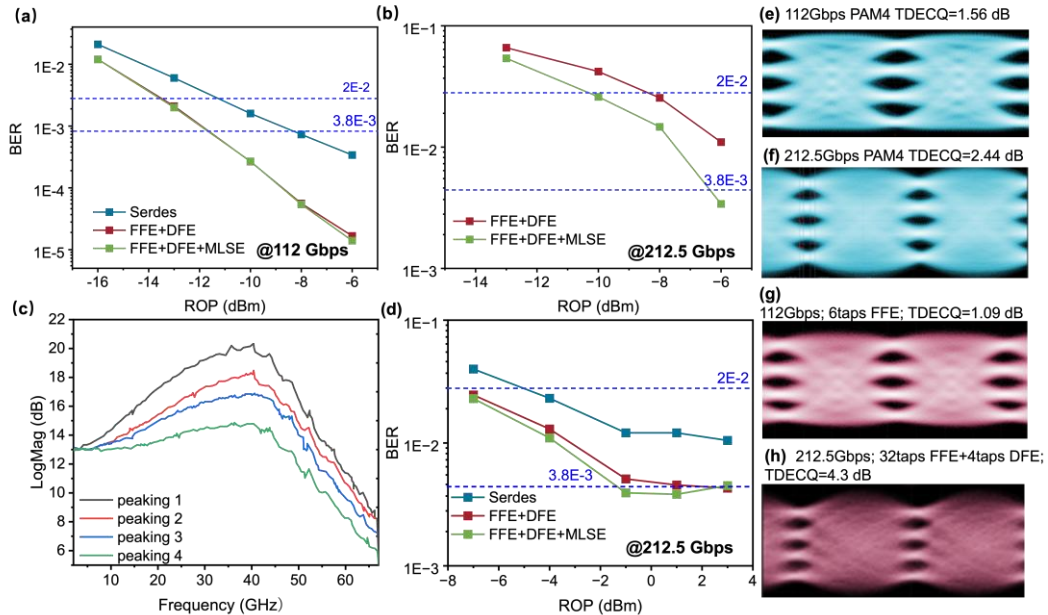


Fig. 3. System link characterization. Experimental BER results for (a) 112 Gbps and (b) 212.5 Gbps PAM4 signals with the SiPh-based platform. (c) S_{21} frequency response of the integrated EIC driver. (d) Experimental BER for 212.5 Gbps PAM4 signals with the TFLN-based platform. (e) 112 Gbps and (f) 212.5 Gbps Back-to-back (B2B) electrical eye diagrams captured at the SerDes receiver. (g) 112 Gbps and 212.5 Gbps (h) the optical eye diagrams output by the SiPh transmitter. ROP: Received Optical Power. FFE:45 taps. DFE:10 taps.

Table 1: Comparison of SiPh and TFLN-based serdes-driven link

Platform	Data Rate (Gbps)	Bandwidth (GHz)	Driver	Receiver Sensitivity (HD-FEC)	Integration
SiPh	212.5 (PAM4)	> 32 (~3 dB, no peaking)	V _{ppd} < 3V, Gain 20 dB, BW(3dB) 53 GHz ~1.1 pJ/bit	-11 dBm	High density
TFLN	212.5 (PAM4)	110 (~6 dB)	Driver-free (Direct from SerDes)	1 dBm	Relatively large

In contrast, the TFLN setup (Fig. 2) adopts a driver-free architecture. A TFLN MZM (EO bandwidth >110 GHz, ~5 dB loss, ~2 V π) is directly driven by the SerDes with 0.5 V_{pp} swing. Eliminating the redriver significantly improves overall link energy efficiency. Specifically, the external driver in the SiPh link consumes ~1.1 pJ/bit; by fully removing this active component, the TFLN direct-drive architecture achieves a substantial reduction in total energy consumption.

Communication Characterization

Figure 3 illustrates the system loopback results of the MZM with 0.5 V_{pp} from SerDes and -3 dB MZM bias.

Figure 3(a) and 3(b) present the BER performance of the SiPh platform. At 112 Gbps, the SiPh link achieves a BER in the order of 1e-4 under direct SerDes detection without compensation. With receiver-side DSP equalization, the BER is improved to below 1e-5. Besides, at 212.5 Gbps, the bandwidth limitations of SiPh MZMs degrade system performance. Here, by leveraging the analog peaking of the EIC driver (as shown in the S₂₁ curve of Fig. 3(c)), combined with DSP equalization, the system BER is successfully maintained below the hard-decision forward error correction (HD-FEC) threshold of 3.8e-3.

For the TFLN platform, Figure 3(d) displays the experimental results under a driver-free architecture. Despite using the same 0.5 V_{pp} drive, the ultra-wideband nature of TFLN (6-dB bandwidth of 110 GHz) allows the system to reach the soft-decision FEC (SD-FEC) threshold even with direct SerDes detection. With offline DSP equalization, signal integrity is further enhanced, bringing the BER well below the HD-FEC threshold.

Figures 3(e) and 3(f) show the captured back-to-back (B2B) electrical eye diagrams at 112 Gbps and 212.5 Gbps, respectively. Additionally, the optical eye diagrams output by the SiPh transmitter are presented in Fig. 3(g) and 3(h).

Table 1 compares the performances of the SiPh-based and TFLN-based optical engines. The SiPh link exhibits better receiver sensitivity, which is primarily attributed to its external driver,

which enhances modulation depth and compensates bandwidth limitations via analog peaking, thereby enhancing the overall signal integrity. In contrast, the driver-free TFLN approach relies on a 0.5 V_{pp} swing from SerDes (with additional cable losses introduced in the experimental setup), resulting in reduced modulation depth and OMA, and thus requires larger input optical power to maintain the target BER. This highlights a key trade-off between driverless and driver-integrated architectures: improved energy efficiency versus reduced optical link budget. Moreover, low-loss co-packaging technologies are essential to mitigate the cable losses in driver-free architectures, enabling energy-efficient solutions without compromising signal integrity.

Conclusion

We demonstrate 212.5 Gbps PAM4 optical interconnects using an LRO architecture, comparing SiPh and TFLN platforms. Through EIC-PIC co-optimization and driver peaking control, SiPh modulators overcome their intrinsic 32 GHz bandwidth limit to achieve a receiver sensitivity of -11 dBm, proving their viability for high-speed, high-density optical interconnects beyond conventional bandwidth constraints. In contrast, the TFLN platform operates driver-free, leveraging its expansive 110 GHz bandwidth for direct SerDes drive to significantly reduce power consumption. However, limited modulation depth degrades sensitivity to 1 dBm, reflecting a trade-off between energy efficiency and optical link budget. This limitation can be further mitigated through low-loss co-packaged optics.

Looking ahead, SerDes-EIC-PIC co-optimization will be critical for scaling SiPh toward 400 Gbps/lane, while driver-free TFLN still remains attractive for ultra-low-power and ultra-high-speed scenarios. Furthermore, advanced low-loss packages will be essential for future linear-drive optical interconnects in AI clusters.

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